

Networks and Integrated Circuits

- To develop the skill to diagnose and rectify the electric circuit networks

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 4491 is the authoritative source for course information, objectives, course outcomes, CO-PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 4491.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Microelectronics
Course code	4491
Course title	Networks and Integrated Circuits
Semester	4 Credits: 1
Course category	Program Core
Periods per week	4 (L:3, T:1, P:0) Periods per semester: 60

Item	Official information
Periods per semester	60
Credits	1
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To develop the skill to diagnose and rectify the electric circuit networks
- To provide an insight on the fundamental concepts of operational amplifiers and their applications.

Course outcomes

CO	Official outcome	Study level
CO1	using the concept of AC signals and the behavior of 13 Applying AC through various components. Apply various network theorems for simplifying	See official syllabus
CO2	electric circuits and networks and illustrate the 15 Applying operations of transformers Illustrate the internal structure, working and 15	See official syllabus
CO3	Applying electrical parameters of operational amplifier.	See official syllabus
CO4	Develop circuits using operational amplifier and Applying timer IC	See official syllabus

CO-PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 2
CO4	CO4 3

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	signals and the behavior of AC through various components.	4	As specified
Module 2	networks and illustrate the operations of transformers.	4	As specified
Module 3	operational amplifier.	4	As specified
Module 4	Develop circuits using operational amplifier and timer IC	5	As specified

MODULE 1

signals and the behavior of AC through various components.

This module is presented from the official syllabus scope for Networks and Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: signals and the behavior of AC through various components.

- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R-

Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R- is an official topic in Module 1 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R- using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the concept of ac signals 2 understanding explain the behavior of ac through r- should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R- means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഇരുപതു Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R- ഇരുപതു ഇരുപതു definition/meaning ഇരുപതു ഇരുപതു. ഇരുപതു ഇരുപതു ഇരുപതു, steps, application ഇരുപതു ഇരുപതു ഇരുപതു. Technical terms English-ഇരുപതു ഇരുപതു ഇരുപതു.

Understanding L-C components 3 Solve the power in given AC circuits by 4

Understanding L-C components 3 Solve the power in given AC circuits by 4 is an official topic in Module 1 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding L-C components 3 Solve the power in given AC circuits by 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding l-c components 3 solve the power in given ac circuits by 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding L-C components 3 Solve the power in given AC circuits by 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 1- Understanding L-C components 3 Solve the power in given AC circuits by 4 definition/meaning
steps, application Technical terms English-

Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4

Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4 is an official topic in Module 1 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying keeping the concept of phasor diagram. solve given series and parallel rlc circuits and 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-ഘാതികാലം അപ്ലൈ ചെയ്തുകൊണ്ട് ഫേസർ ഡയഗ്രാം. Solve given series and parallel RLC circuits and 4 ഘാതികാലം ഫേസർ ഡയഗ്രാം definition/meaning ഘാതികാലം ഫേസർ ഡയഗ്രാം, steps, application ഘാതികാലം ഫേസർ ഡയഗ്രാം ഫേസർ. Technical terms English-ഘാതികാലം ഫേസർ ഡയഗ്രാം.

Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and

Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and is an official topic in Module 1 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying find the various parameters triangular and sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor ac through resistance, inductance, and capacitance (solve simple problems) - phasor diagrams- power factor definition - calculation of active, reactive and apparent power, resonance and q factor in an rlc circuit - series and parallel ac circuits (simple problems). apply various network theorems for simplifying electric circuits and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

[illegible]

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

signals and the behavior of AC through various components.

M1.01 Illustrate the concept of AC signals 2
Understanding

Explain the behavior of AC through R-

M1.02 L-C components 3
Understanding

Solve the power in given AC circuits by 4

M1.03 keeping the concept of phasor diagram. 4
Applying
Solve given series and parallel RLC circuits and

M1.04 find the various parameters
Applying

Contents: Concept of alternating voltage and current - Different waveforms (Sine, Square, Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

networks and illustrate the operations of transformers.

This module is presented from the official syllabus scope for Networks and Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: networks and illustrate the operations of transformers.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Understanding Apply various network theorems for solving 6

Understanding Apply various network theorems for solving 6 is an official topic in Module 2 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding Apply various network theorems for solving 6 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding apply various network theorems for solving 6 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding Apply various network theorems for solving 6 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-Understanding Apply various network theorems for solving 6 problems. definition/meaning. steps, application. Technical terms English- Malayalam.

Applying electrical and electronics circuits. Explain principle and operations of transformers 3

Applying electrical and electronics circuits. Explain principle and operations of transformers 3 is an official topic in Module 2 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Applying electrical and electronics circuits. Explain principle and operations of transformers 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, applying electrical and electronics circuits. explain principle and operations of transformers 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Applying electrical and electronics circuits. Explain principle and operations of transformers 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-എപ്ലൈംഗ് ഇലക്ട്രിക്കൽ ആൻഡ് ഇലക്ട്രോണിക്സ് സർക്യൂട്ട്സ്. Explain principle and operations of transformers 3 ഏകദേശം 300 വാക്കുകൾ. definition/meaning 100 വാക്കുകൾ. ഏകദേശം 100 വാക്കുകൾ, steps, application 100 വാക്കുകൾ. Technical terms English-ൽ എഴുതുക. 100 വാക്കുകൾ.

Understanding List the types and applications of Transformers. 3

Understanding List the types and applications of Transformers. 3 is an official topic in Module 2 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding List the types and applications of Transformers. 3 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding list the types and applications of transformers. 3 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding List the types and applications of Transformers. 3 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2- Understanding List the types and applications of Transformers. 3 ഉപതരസ്സങ്ങൾ തരസ്സം definition/meaning ഉപതരസ്സങ്ങൾ. ഉപതരസ്സം ഉപതരസ്സം, steps, application ഉപതരസ്സം ഉപതരസ്സം ഉപതരസ്സം. Technical terms English- ഉപതരസ്സം ഉപതരസ്സം.

Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of

Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of is an official topic in Module 2 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, remembering ohm's law - kirchhoff's law - mesh analysis - node analysis- superposition theorem - thevenin's theorem - maximum power transfer theorem (solve simple problems) transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers illustrate the internal structure, working and electrical parameters of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2- Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of definition/meaning . steps, application . Technical terms English- .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

networks and illustrate the operations of transformers.	
Illustrate various network theorems	3
M2.01	
Understanding	
Apply various network theorems for solving	6
M2.02	
Applying	
electrical and electronics circuits.	
Explain principle and operations of transformers	3
M2.03	
Understanding	
List the types and applications of Transformers.	3
M2.04	
Remembering	
Series Test - I	1
Contents: Network Theorems and Transformers	
Ohm's law - Kirchhoff's law – Mesh analysis – Node analysis- Superposition theorem -	
Thevenin's theorem - Maximum power transfer theorem (Solve simple problems)	
Transformers: working principle of transformer - construction of transformer - elementary	
theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf	
equation derivation - losses in transformers -	
types, applications of transformers	

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

operational amplifier.

This module is presented from the official syllabus scope for Networks and Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: operational amplifier.
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Illustrate the operation of differential amplifier

Illustrate the operation of differential amplifier is an official topic in Module 3 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate the operation of differential amplifier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate the operation of differential amplifier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate the operation of differential amplifier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-
Illustrate the operation of differential amplifier
definition/meaning .
steps, application . Technical terms English-
.

Explain the block diagram of operational amplifier

Explain the block diagram of operational amplifier is an official topic in Module 3 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the block diagram of operational amplifier using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the block diagram of operational amplifier should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the block diagram of operational amplifier means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- Explain the block diagram of operational amplifier
 definition/meaning .
 , steps, application . Technical terms English-
 .

Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and

Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and is an official topic in Module 3 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain operational amplifier parameters 3 understanding illustrate the basic op-amp feedback circuits and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-ഓപ്പറേഷണൽ അംപ്ലിഫയർ പാരാമീറ്ററുകൾ 3 Understanding Illustrate the basic op-amp feedback circuits and ഓപ്പറേഷണൽ അംപ്ലിഫയർ definition/meaning ഓപ്പറേഷണൽ അംപ്ലിഫയർ. ഓപ്പറേഷണൽ അംപ്ലിഫയർ steps, application ഓപ്പറേഷണൽ അംപ്ലിഫയർ പാരാമീറ്ററുകൾ. Technical terms English-ഓപ്പറേഷണൽ അംപ്ലിഫയർ.

6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier.

6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier. is an official topic in Module 3 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier. using standard technical terminology.

- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 6 applying develop various amplifier circuits. fundamentals of operational amplifier differential amplifier: bjt differential amplifier - types -large and small signal operation- input resistance - voltage gain - cmrr - non-ideal characteristics operational amplifier: op-amp symbol - package types - block diagram - dc and ac op-amp parameters - equivalent circuit - voltage transfer curve - open loop op-amp configurations - op-amp with negative feedback - properties of practical op-amp - inverting amplifier - non-inverting amplifier - voltage follower - summing amplifier - difference amplifier - scaling and averaging amplifiers - adder - subtractor - instrumentation amplifier. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3- 6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package

types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder – subtractor - Instrumentation amplifier.
 definition/meaning . , steps, application . Technical terms English- . .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

operational amplifier.

M3.01 Illustrate the operation of differential amplifier 4
Understanding

M3.02 Explain the block diagram of operational amplifier 2
Understanding

M3.03 Explain operational amplifier parameters 3
Understanding

M3.04 Illustrate the basic op-amp feedback circuits and 6
Applying
develop various amplifier circuits.

Contents:

Fundamentals of operational amplifier

Differential amplifier: BJT differential amplifier - types -Large and small signal operation-

Input resistance - Voltage gain - CMRR - non-ideal characteristics

Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op-amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier -

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam

MODULE 4

Develop circuits using operational amplifier and timer IC

This module is presented from the official syllabus scope for Networks and Integrated Circuits. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Develop circuits using operational amplifier and timer IC
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

Develop wave-shaping circuits using op-amp

Develop wave-shaping circuits using op-amp is an official topic in Module 4 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop wave-shaping circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop wave-shaping circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop wave-shaping circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓപ്പറേഷൻ അംപ് വേവ്-ഷാപ്പിംഗ് സിരക്യൂട്ടുകൾ ഉപയോഗിച്ച് വികസിപ്പിക്കുക. ഓപ്പറേഷൻ അംപ് വേവ്-ഷാപ്പിംഗ് സിരക്യൂട്ടുകൾ, definition/meaning, steps, application, application ഓപ്പറേഷൻ അംപ് വേവ്-ഷാപ്പിംഗ് സിരക്യൂട്ടുകൾ. Technical terms English-ഓപ്പറേഷൻ അംപ് വേവ്-ഷാപ്പിംഗ് സിരക്യൂട്ടുകൾ.

Develop oscillator circuits using op-amp

Develop oscillator circuits using op-amp is an official topic in Module 4 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop oscillator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop oscillator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop oscillator circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓസിലേറ്റർ സർക്യൂട്ട് വികസിപ്പിക്കുക. ഓസിലേറ്റർ സർക്യൂട്ട് വികസിപ്പിക്കുക, steps, application ഓസിലേറ്റർ സർക്യൂട്ട് വികസിപ്പിക്കുക. Technical terms English-ഓസിലേറ്റർ സർക്യൂട്ട് വികസിപ്പിക്കുക.

Illustrate comparator circuits using op-amp

Illustrate comparator circuits using op-amp is an official topic in Module 4 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Illustrate comparator circuits using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, illustrate comparator circuits using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Illustrate comparator circuits using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓപ് അംപ് ഉപയോഗിച്ച് കമ്പാരേറ്റർ സർക്യൂട്ടുകൾക്ക് ഉദാഹരണം നൽകുക. ഓപ് അംപ് കമ്പാരേറ്റർ സർക്യൂട്ടിന്റെ നിർവ്വചനം/അർത്ഥം, ഓപ് അംപ് കമ്പാരേറ്റർ സർക്യൂട്ടിന്റെ ഓപറേറ്റിംഗ് സ്റ്റേപ്പുകൾ, ഓപ് അംപ് കമ്പാരേറ്റർ സർക്യൂട്ടിന്റെ പ്രയോഗങ്ങൾ. Technical terms English-ഓപ് അംപ് കമ്പാരേറ്റർ സർക്യൂട്ട്.

Develop active filters using op-amp

Develop active filters using op-amp is an official topic in Module 4 of Networks and Integrated Circuits. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop active filters using op-amp using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop active filters using op-amp should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop active filters using op-amp means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-ഓപ് അംപ് ഉപയോഗിച്ച് ആക്റ്റീവ് ഫിൽറ്ററുകൾ വികസിപ്പിക്കുക

ഓപ് അംപ് ഫിൽറ്ററുകളുടെ നിർവ്വചന/അർത്ഥം, ഓപ് അംപ് ഫിൽറ്ററുകളുടെ ഘട്ടങ്ങൾ, പ്രയോഗം എന്നിവയെക്കുറിച്ചുള്ള വിവരങ്ങൾ. Technical terms English-ൽ നിന്ന് മലയാളത്തിലേക്ക്.

Develop circuits using 555 timer IC 3 Applying Text/Reference T/R Book
Title/Author B. L. Thereja and A. K. Thereja, "Textbook of Electrical
Technology: Part 1 T1 - Basic Electrical Engineering in S. I. Units", S. Chand
Publication, 2012 Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age
International, T2 3/e, 2010 Sergio Franco, Design with operational amplifiers
and analog integrated T3 circuits, 3rd Edition, Tata McGraw-Hill, 2008 B. L.
Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 2-AC R1
and DC machines", S. Chand Publication, 2012 Dr Ganesh Rao& R Sreenivasa,
"Network Analysis: A simplified approach", R2 Cengage Gayakwad R. A., Op-
Amps and Linear Integrated Circuits, Prentice Hall, R3 4/e, 2010 R4 Botkar K.
R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K.
Bhaaskaran, Linear Integrated Circuits, Tata R5 McGraw Hill, 2008 David A.
Bell, Operational Amplifiers & Linear ICs, Oxford University R6 Press, 2nd
edition,2010 Online resources: Sl.No. Website Link 1
<https://www.khanacademy.org/electrical/> 2 <https://ocw.mit.edu/electrical> 3
<https://nptel.ac.in/courses/117107094/> 4
<https://nptel.ac.in/courses/117101106/> 5
<https://nptel.ac.in/courses/108/108/108108111/> 6
<https://inst.eecs.berkeley.edu/~ee140/fa19/> 7
<https://freevideolectures.com/course/2915/linear-integrated-circuits>

Develop circuits using 555 timer IC 3 Applying Text/Reference T/R Book
 Title/Author B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology:
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 Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T2 3/e,
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 "Textbook of Electrical Technology: Part 2-AC R1 and DC machines", S. Chand
 Publication, 2012 Dr Ganesh Rao& R Sreenivasa, "Network Analysis: A simplified
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 Oxford University R6 Press, 2nd edition,2010 Online resources: Sl.No. Website
 Link 1 <https://www.khanacademy.org/electrical/> 2 <https://ocw.mit.edu/electrical> 3
<https://nptel.ac.in/courses/117107094/> 4 <https://nptel.ac.in/courses/117101106/> 5
<https://nptel.ac.in/courses/108/108/108108111/> 6
<https://inst.eecs.berkeley.edu/~ee140/fa19/> 7
<https://freevideolectures.com/course/2915/linear-integrated-circuits> is an official
 topic in Module 4 of Networks and Integrated Circuits. Study the terminology first,

then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Develop circuits using 555 timer IC 3 Applying Text/Reference T/R Book Title/Author B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 1 T1 - Basic Electrical Engineering in S. I. Units”, S. Chand Publication, 2012 Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T2 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T3 circuits, 3rd Edition, Tata McGraw-Hill, 2008 B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 2-AC R1 and DC machines”, S. Chand Publication, 2012 Dr Ganesh Rao& R Sreenivasa, “Network Analysis: A simplified approach”, R2 Cengage Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R3 4/e, 2010 R4 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 SalivahananS. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R5 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R6 Press, 2nd edition,2010 Online resources: Sl.No. Website Link 1 <https://www.khanacademy.org/electrical/> 2 <https://ocw.mit.edu/electrical> 3 <https://nptel.ac.in/courses/117107094/> 4 <https://nptel.ac.in/courses/117101106/> 5 <https://nptel.ac.in/courses/108/108/108108111/> 6 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 7 <https://freevideolectures.com/course/2915/linear-integrated-circuits> using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, develop circuits using 555 timer ic 3 applying text/reference t/r book title/author b. l. thereja and a. k. thereja, “textbook of electrical technology: part 1 t1 - basic electrical engineering in s. i. units”, s. chand publication, 2012 roy d. c. and s. b. jain, linear integrated circuits, new age international, t2 3/e, 2010 sergio franco, design with operational amplifiers and analog integrated t3 circuits, 3rd edition, tata mcgraw-hill, 2008 b. l. thereja and a. k. thereja, “textbook of electrical technology: part 2-ac r1 and dc machines”, s. chand publication, 2012 dr ganesh rao& r sreenivasa,

“network analysis: a simplified approach”, r2 cengage gayakwad r. a., op-amps and linear integrated circuits, prentice hall, r3 4/e, 2010 r4 botkar k. r., integrated circuits, 10/e, khanna publishers, 2010 salivahanans. ,v. s. k. bhaaskaran, linear integrated circuits, tata r5 mcgraw hill, 2008 david a. bell, operational amplifiers & linear ics, oxford university r6 press, 2nd edition,2010 online resources: sl.no. website link 1 <https://www.khanacademy.org/electrical/> 2 <https://ocw.mit.edu/electrical> 3 <https://nptel.ac.in/courses/117107094/> 4 <https://nptel.ac.in/courses/117101106/> 5 <https://nptel.ac.in/courses/108/108/108108111/> 6 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 7 <https://freevideolectures.com/course/2915/linear-integrated-circuits> should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Develop circuits using 555 timer IC 3 Applying Text/Reference T/R Book Title/Author B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 1 T1 - Basic Electrical Engineering in S. I. Units”, S. Chand Publication, 2012 Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T2 3/e, 2010 Sergio Franco, Design with operational amplifiers and analog integrated T3 circuits, 3rd Edition, Tata McGraw-Hill, 2008 B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 2-AC R1 and DC machines”, S. Chand Publication, 2012 Dr Ganesh Rao& R Sreenivasa, “Network Analysis: A simplified approach”, R2 Cengage Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R3 4/e, 2010 R4 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 Salivahanans. ,V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R5 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R6 Press, 2nd edition,2010 Online resources: Sl.No. Website Link 1 https://www.khanacademy.org/electrical/ 2 https://ocw.mit.edu/electrical 3 https://nptel.ac.in/courses/117107094/ 4 https://nptel.ac.in/courses/117101106/ 5 https://nptel.ac.in/courses/108/108/108108111/ 6 https://inst.eecs.berkeley.edu/~ee140/fa19/ 7 https://freevideolectures.com/course/2915/linear-integrated-circuits means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ Develop circuits using 555 timer IC 3 Applying Text/Reference T/R Book Title/Author B. L. Thereja and A. K. Thereja, “Textbook of Electrical Technology: Part 1 T1 - Basic Electrical Engineering in S. I. Units”, S. Chand Publication, 2012 Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age International, T2 3/e, 2010 Sergio Franco, Design with operational amplifiers

and analog integrated T3 circuits, 3rd Edition, Tata McGraw-Hill, 2008 B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 2-AC R1 and DC machines", S. Chand Publication, 2012 Dr Ganesh Rao & R Sreenivasa, "Network Analysis: A simplified approach", R2 Cengage Gayakwad R. A., Op-Amps and Linear Integrated Circuits, Prentice Hall, R3 4/e, 2010 R4 Botkar K. R., Integrated Circuits, 10/e, Khanna Publishers, 2010 Salivahanan S. , V. S. K. Bhaaskaran, Linear Integrated Circuits, Tata R5 McGraw Hill, 2008 David A. Bell, Operational Amplifiers & Linear ICs, Oxford University R6 Press, 2nd edition, 2010 Online resources: Sl.No. Website Link 1 <https://www.khanacademy.org/electrical/> 2 <https://ocw.mit.edu/electrical> 3 <https://nptel.ac.in/courses/117107094/> 4 <https://nptel.ac.in/courses/117101106/> 5 <https://nptel.ac.in/courses/108/108/108108111/> 6 <https://inst.eecs.berkeley.edu/~ee140/fa19/> 7 <https://freevideolectures.com/course/2915/linear-integrated-circuits>
 definition/meaning . steps, application . Technical terms English-
 .

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module.
Educational explanations below are separate elaboration.

Develop circuits using operational amplifier and timer IC

M4.01	Develop wave-shaping circuits using op-amp	3	
Applying			
M4.02	Develop oscillator circuits using op-amp	3	Applying
M4.03	Illustrate comparator circuits using op-amp	3	Applying
M4.04	Develop active filters using op-amp	3	Applying
M4.05	Develop circuits using 555 timer IC	3	Applying
	Series Test – II	1	

Text/Reference

T/R

Book Title/Author

B. L. Thereja and A. K. Thereja, "Textbook of Electrical Technology: Part 1
T1 - Basic Electrical Engineering in S. I. Units", S. Chand Publication, 2012

Roy D. C. and S. B. Jain, Linear Integrated Circuits, New Age
International,
T2 3/e, 2010

Sergio Franco, Design with operational amplifiers and analog integrated
T3 circuits, 3rd Edition, Tata McGraw-Hill, 2008

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

Definition → Principle → Components/steps → Application → Conclusion

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

Aim → Apparatus → Procedure → Observation → Result → Safety

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

The official assessment structure is reproduced from the supplied PDF.

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R-. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R-*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Understanding L-C components 3 Solve the power in given AC circuits by 4. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding L-C components 3 Solve the power in given AC circuits by 4*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4. (3 marks)

Answer: Begin with the standard definition or identity of *Applying keeping the concept of phasor diagram. Solve given series and parallel RLC circuits and 4*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and. (3 marks)

Answer: Begin with the standard definition or identity of *Applying find the various parameters Triangular and Sawtooth) - representation of alternating quantities - define*

the terms cycle, time period, frequency, amplitude, phase, maximum value, rms value, average value form factor AC through resistance, inductance, and capacitance (Solve simple problems) - Phasor diagrams- power factor definition - calculation of active, reactive and apparent power, Resonance and Q factor in an RLC circuit - series and parallel AC circuits (simple problems). Apply various network theorems for simplifying electric circuits and. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Module 2

Define or explain Understanding Apply various network theorems for solving 6. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding Apply various network theorems for solving 6*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Applying electrical and electronics circuits. Explain principle and operations of transformers 3. (3 marks)

Answer: Begin with the standard definition or identity of *Applying electrical and electronics circuits. Explain principle and operations of transformers 3*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Understanding List the types and applications of Transformers. 3. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding List the types and applications of Transformers. 3.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of. (3 marks)

Answer: Begin with the standard definition or identity of *Remembering Ohm's law - Kirchhoff's law - Mesh analysis - Node analysis- Superposition theorem - Thevenin's theorem - Maximum power transfer theorem (Solve simple problems) Transformers: working principle of transformer - construction of transformer - elementary theory of an ideal transformer - voltage transformation ratio and rating of a transformer - emf equation derivation - losses in transformers - types, applications of transformers Illustrate the internal structure, working and electrical parameters of.* State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 3

Define or explain Illustrate the operation of differential amplifier. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate the operation of differential amplifier*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain the block diagram of operational amplifier. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the block diagram of operational amplifier*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain operational amplifier parameters 3 Understanding Illustrate the basic op-amp feedback circuits and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain 6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-

inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier.. (3 marks)

Answer: Begin with the standard definition or identity of 6 Applying develop various amplifier circuits. Fundamentals of operational amplifier Differential amplifier: BJT differential amplifier - types -Large and small signal operation- Input resistance - Voltage gain - CMRR - non-ideal characteristics Operational Amplifier: Op-amp symbol - package types - block diagram - DC and AC op- amp parameters - equivalent circuit - voltage transfer curve - Open loop op-amp configurations - Op-amp with negative feedback - Properties of practical op-amp - Inverting amplifier - Non-inverting amplifier - Voltage follower - Summing amplifier - difference amplifier - Scaling and averaging amplifiers - adder - subtractor - Instrumentation amplifier.. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Module 4

Define or explain Develop wave-shaping circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Develop wave-shaping circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□ □□□□□□.

Define or explain Develop oscillator circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Develop oscillator circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Illustrate comparator circuits using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Illustrate comparator circuits using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

Define or explain Develop active filters using op-amp. (3 marks)

Answer: Begin with the standard definition or identity of *Develop active filters using op-amp*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□ □□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Illustrate the concept of AC signals 2 Understanding Explain the behavior of AC through R-**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **Understanding Apply various network theorems for solving 6.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **Illustrate the operation of differential amplifier.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **Develop wave-shaping circuits using op-amp.**

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.
- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No. Website Link <https://www.khanacademy.org/electrical/>
<https://ocw.mit.edu/electrical> <https://nptel.ac.in/courses/117107094/>
<https://nptel.ac.in/courses/117101106/>
- <https://nptel.ac.in/courses/108/108/108108111/>
<https://inst.eecs.berkeley.edu/~ee140/fa19/>
- <https://freevideolectures.com/course/2915/linear-integrated-circuits>

IN-PAGE SEARCH

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